Chapter 2 (Part 2): Auctions, Iterated Dominance, & Nash Equilibrium

ECON 5001
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Based on: *Game Theory* by Giacomo Bonanno

9/16

Updated 2026-09-10 at 00:55:02

Where We Left Off

We can now:

- separate a **frame** from a **game**
- write preferences as a complete, transitive $\succsim_i$
- represent $\succsim_i$ with an ordinal utility function
- compare two strategies of one player: **dominance**

And we found: if you have a dominant strategy, play it.

In this lecture: what if you don't?

Log In: Auction Game

- MobLab: Two auctions. You're bidders.
- You'll be told what the item is worth **to you**
  - That number is yours alone; others have different ones
  - Random from \$10 to \$100 (equally likely)
- Groups of SIX bidders.
- Submit one sealed bid. Highest bid wins.
  - First game: Winner pays their bid
  - Second game: Winner pays highest *losing* bid (2nd highest bid)
- We'll play it **twice**, under two different rules

Your payoff is your value minus what you pay, or zero if you don't win.

No talking. Bid however you think is smart.

What Did You Just Play?

Round 1, first-price auction: the winner pays *their own bid*.

Round 2, second-price auction: the winner pays *the 2nd highest bid*.

In which round did you bid closer to your true value?
Did you (bid $<$ value) or (bid $>$ value)?

Second-Price Auctions

The Rules

A **second-price** (or **Vickrey**) auction:

- sealed bids: nobody sees anyone else's bid
- highest bidder wins
- the winner pays the **second-highest** bid, not their own
- ties are broken by a rule fixed in advance

Why would a seller ever do this? It looks like giving money away.

An Example: The Game-Day Permit

Suppose OSU auctions off a food truck permit near the stadium for game day. Dos Hermanos and Pitabilities both want it.

- Dos Hermanos (Player 1): estimates **$30k** extra profit if they get it
  This is their **value** for the permit. Write $v_1 = 30$ (in thousands)
- Suppose bids must be one of $10k, $20k, $30k, $40k, $50k, $60k
  So $S_1 = S_2 = \{10, 20, 30, 40, 50, 60\}$. They have 5 possible strategies. ← strategy sets
  bids → Write bids as $b_1 \in S_1$ and $b_2 \in S_2$
- **Second-price auction:** highest bid wins, pays *2nd highest bid*
- Ties go to Pitabilities (decided by a coin flip beforehand) ← arbitrary

Dos Hermanos's payoff in the Second Price Auction:

your value → other's bid

$$\text{payoff}: \quad \pi_1 = \begin{cases} v_1 - b_2 & \text{if } b_1 > b_2 \leftarrow \text{win} \\ 0 & \text{if } b_1 \leq b_2 \leftarrow \text{lose} \end{cases}$$

Assume utility = money (selfish & greedy)

Only Dos Hermanos's Payoffs Shown

Player 1

Red = lose

Green = win

Dos Hermanos ($v_1 = 30$) \ Pitabilities	10	20	30	40	50	60
10	0, ·	0, ·	0, ·	0, ·	0, ·	0, ·
20	20, ·	0, ·	0, ·	0, ·	0, ·	0, ·
30	20, ·	10, ·	0, ·	0, ·	0, ·	0, ·
40	20, ·	10, ·	0, ·	0, ·	0, ·	0, ·
50	20, ·	10, ·	0, ·	−10, ·	0, ·	0, ·
60	20, ·	10, ·	0, ·	−10, ·	−20, ·	0, ·

your earnings doesn't depend on your bid

weakly dom

equil.

weakly dominated by 30 and 40

30 − 50

Recall: $\pi_1 = \begin{cases} v_1 - b_2 & \text{if } b_1 > b_2 \\ 0 & \text{if } b_1 \leq b_2 \end{cases}$ with $v_1 = 30$

Reading the Table

- $b_1 = 30$ is a ___weakly dominant___ strategy

bid = value

- $b_1 = 40$ is a ___weakly dominant___ strategy

bid just above value

- Strategies $b_1 = 30$ and $b_1 = 40$ are ___equivalent___

- Overbidding ($b_1 = 50$) is weakly dominated by $b_1 = 30$. Where is it strictly worse?

when $b_1 > b_2 > V_1$ you pay more than it's worth

- Underbidding ($b_1 = 20$) is weakly dominated by $b_1 = 30$. Where is it strictly worse?

when $V_1 > b_2 > b_1$ you lose

$b_1 = V_1 > b_2$ would make money

Vickrey's Theorem

Theorem (Vickrey, 1961). In a second-price auction, if player i is selfish and greedy, then bidding their true value ($b_i = v_i$) is a **weakly dominant strategy**.

Remarkable, because:

- it holds no matter how many bidders there are
- it holds no matter what anyone else bids
- bidders don't need to guess, predict, or model their rivals at all!

all very simple
just bid your value regardless

Lab result: many people don't realize this, end up overbidding ($b_i > v_i$)
"Failure of contingent thinking"

Why It Works: Two Cases

Let v_1 be my value and b_2 the other bid. Compare bidding v_1 to anything else.

Case 1: $b_2 \leq v_1$. Bidding v_1 wins and pays b_2, for a payoff of $v_1 - b_2 \geq 0$.

- Bidding higher: still win, still pay b_2. ____________________
- Bidding below b_2: lose, payoff 0. ____________________

Case 2: $b_2 > v_1$. Bidding v_1 loses, for a payoff of 0.

- Bidding anything below b_2: still lose. ____________________
- Bidding at or above b_2: win, pay b_2. ____________________

Raising your bid can only ever change *whether* you win, never *what you pay*.

The Fine Print

The theorem assumes the bidder is **selfish and greedy**.

Drop that and it fails. Suppose I also care what *you* pay:

- **Spiteful**: if I lose, I'd rather you paid more

  ↑ bid

- **Benevolent**: if I lose, I'd rather you paid less

  ↓ bid

In either case, bidding my value is no longer dominant.

Again: *preferences matter!*

The Pivotal Mechanism

The Pivotal Mechanism

I'm skipping this due to time.
This will **not** appear on a quiz or exam.
In a nutshell:

- We're thinking about building a park.
- Person i's value: v_i. Total cost: C
- Social Welfare: $W = \sum_i v_i - C$ ← total value − cost
- Pareto optimal: Build if $W \geq 0$ not if $W < 0$
- Suppose your tax is based on v_i. You'd lie about v_i!!

The fix: the **pivotal** mechanism

- Social Welfare if i didn't exist: W_{-i}
- Player i is **pivotal** if $W_{-i} < 0$ but $W \geq 0$ (or vice-versa)
- Player i's tax is W_{-i}
- **Result:** It's a weakly dom. strategy to report v_i truthfully

Exactly like a 2nd-price auction!!

Intellectual History

Vickrey, 1961

Clarke, 1971

Groves, 1973

- **1973**: Groves publishes the pivotal mechanism
- **1971**: Clarke had already published a special case
- **1961**: People realized it's just like Vickrey's 2nd price auction

The "pivotal mechanism" is often called "the **VCG** mechanism."

Vickrey won the Nobel Prize in 1996 but died three days after it was announced.

Next Solution Concept:
Iterated Deletion of Dominated Strategies

An Example

Player 1 \ Player 2	L	C	R
T	$\underline{4}$, $\underline{3}$	5, 1	$\underline{6}$, 2
M	2, 1	8, 4	3, $\underline{6}$
B	3, 0	$\underline{9}$, 6	2, $\underline{8}$

1. Underline best-response payoffs

No dom. strategy

An Example

		Player 2		
		L	C	R
Player 1	T	4, 3	5, 1	6, 2
	M	2, 1	8, 4	3, 6
	B	3, 0	9, 6	2, 8

1. Underline best-response payoffs
2. There are no dominant strategies. But are any *strictly dominated*?
   - Fact: dominated ⇒ never a best response (not underlined)
   - So, check *M* and *C*
3. *C* is strictly dominated by __R____. Delete it!!
   - Player 1 knows Player 2 will **not** play *C*

An Example

		Player 2		
		L	C	R
Player 1	T	4, 3	5, 1	6, 2
	M	2, 1	8, 4	3, 6
	B	3, 0	9, 6	2, 8

1. Underline best-response payoffs
2. There are no dominant strategies. But are any *strictly dominated*?
    - Fact: dominated $\Rightarrow$ never a best response (not underlined)
    - So, check *M* and *C*
3. *C* is strictly dominated by ________. Delete it!!
    - Player 1 knows Player 2 will **not** play *C*
4. But now *M* and *B* are strictly dominated by __T__. Delete them

An Example

		Player 2		
		L	C	R
Player 1	T	$\underline{4}, \underline{3}$	5, 1	$\underline{6}$, 2
	M	2, 1	8, 4	3, $\underline{6}$
	B	3, 0	$\underline{9}$, 6	2, $\underline{8}$

Predict
(T, L)

1. Underline best-response payoffs
2. There are no dominant strategies. But are any *strictly dominated*?
   - Fact: dominated $\Rightarrow$ never a best response (not underlined)
   - So, check *M* and *C*
3. *C* is strictly dominated by ________. Delete it!!
   - Player 1 knows Player 2 will **not** play *C*
4. But now *M* and *B* are strictly dominated by ________. Delete them
5. Now *R* is dominated by *L*. Delete *R*!

An Example

Player 1 \ Player 2	L	C	R
T	4, 3	5, 1	6, 2
M	2, 1	8, 4	3, 6
B	3, 0	9, 6	2, 8

1. Underline best-response payoffs
2. There are no dominant strategies. But are any *strictly dominated*?
    - Fact: dominated $\Rightarrow$ never a best response (not underlined)
    - So, check M and C
3. C is strictly dominated by ________. Delete it!!
    - Player 1 knows Player 2 will **not** play C
4. But now M and B are strictly dominated by ________. Delete them
5. Now R is dominated by L. Delete R!
6. Our solution: $s^* = (T, L)$

Our new solution concept:

Iterated
Deletion of
Strictly
Dominated
Strategies

IDSDS

(Also known as **IE**SDS, for "Elimination" instead of "Deletion")

Iterated Deletion of Strictly Dominated Strategies.

$G =$

	L	C	R
T			
m			
B			

Start with game G. ← original game

- G^1: delete every strictly dominated strategy, for every player
- G^2: do it again, inside G^1
- ⋮
- G^∞: the output, reached in finitely many steps if G is finite

G' "smaller" than G

G':

	L	R
T		
m		
B		

G^2:

	L	R
T		

If G^∞ is a single strategy profile, we call it the **iterated strict dominant-strategy profile.** Otherwise it's just "the output of IDSDS."

IDSDS solution

G^3:

	L
T	

single s^*

$G^\infty = G^3$

G^∞ is not a single profile/cell then IDSDS is inconclusive

A Rationality Logic

We can justify IDSDS via "common knowledge of rationality"

eliminate strictly dom strats

1. I know you're rational $\Rightarrow$ you won't play a strictly dominated strategy. G^1
2. I know you know I'm rational $\Rightarrow$ I know you'll eliminate my remaining strictly dominated strategies. G^2
3. I know you know I know you're rational $\Rightarrow$ I know you know that I'll eliminate your remaining strictly dominated strategies. G^3

   ...

∞. I know that you know that I know that... $\Rightarrow$ IDSDS. G^∞

Downside: may not give a unique prediction

Upside: good foundations in terms of rationality

Shortcoming of IDSDS

Player 1 \ Player 2	W	X	Y	Z
A	3, 4	6, 6	2, 1	2, 0
B	2, 1	1, 2	5, 5	2, 3
C	7, 3	3, 4	1, 3	2, 2
D	2, 8	1, 1	1, 2	9, 0

1. Underline best-response payoffs

Shortcoming of IDSDS

never BR
strictly dominated?

YES

Player 2

Player 1	W	X	Y	Z
A	3, 4	6, 6	2, 1	2, 0
B	2, 1	1, 2	5, 5	2, 3
C	7, 3	3, 4	1, 3	2, 2
D	2, 8	1, 1	1, 2	9, 0

1. Underline best-response payoffs
2. No dominant strategies. But are any *strictly dominated*?
    - Dominated $\Rightarrow$ never a best response. Only Z has no underline
3. Z is strictly dominated by Y. Delete it!

Shortcoming of IDSDS

Player 2

		W	X	Y	Z
Player 1	A	3, 4	6, 6	2, 1	2, 0
	B	2, 1	1, 2	5, 5	2, 3
	C	7, 3	3, 4	1, 3	2, 2
	D	2, 8	1, 1	1, 2	9, 0

never BR

1. Underline best-response payoffs
2. No dominant strategies. But are any *strictly dominated*?
    - Dominated ⇒ never a best response. Only *Z* has no underline
3. *Z* is strictly dominated by ______. Delete it!
4. *D*'s only underline was in *Z*. Now it is dominated by A. Delete it!

Shortcoming of IDSDS

never BR

Player 2

Player 1	~~W~~	X	Y	~~Z~~
A	3, 4	$\underline{6}$, $\underline{6}$	2, 1	~~2, 0~~
B	2, 1	1, 2	$\underline{5}$, $\underline{5}$	~~2, 3~~
C	$\underline{7}$, 3	3, $\underline{4}$	1, 3	~~2, 2~~
~~D~~	~~2, $\underline{8}$~~	~~1, 1~~	~~1, 2~~	~~$\underline{9}$, 0~~

1. Underline best-response payoffs
2. No dominant strategies. But are any *strictly dominated*?
   - Dominated $\Rightarrow$ never a best response. Only *Z* has no underline
3. *Z* is strictly dominated by _______. Delete it!
4. *D*'s only underline was in *Z*. Now it is dominated by _______. Delete it!
5. *W*'s only underline was in *D*. Now it is dominated by X. Delete it!

Shortcoming of IDSDS

		Player 2			
		W	X	Y	Z
Player 1	A	3, 4	$\underline{6}$, $\underline{6}$	2, 1	2, 0
	B	2, 1	1, 2	$\underline{5}$, $\underline{5}$	2, 3
	C	$\underline{7}$, 3	3, $\underline{4}$	1, 3	2, 2
	D	2, $\underline{8}$	1, 1	1, 2	$\underline{9}$, 0

never BR

dominated?

1. Underline best-response payoffs
2. No dominant strategies. But are any *strictly dominated*?
    - Dominated $\Rightarrow$ never a best response. Only *Z* has no underline
3. *Z* is strictly dominated by _______. Delete it!
4. *D*'s only underline was in *Z*. Now it is dominated by _______. Delete it!
5. *W*'s only underline was in *D*. Now it is dominated by _______. Delete it!
6. *C*'s only underline was in *W*. Now it is dominated by A. Delete it!

Shortcoming of IDSDS

Player 2

Player 1

	W	X	Y	Z
A	3, 4	$\underline{6}$, $\underline{6}$	2, 1	2, 0
B	2, 1	1, 2	$\underline{5}$, $\underline{5}$	2, 3
C	$\underline{7}$, 3	3, $\underline{4}$	1, 3	2, 2
D	2, $\underline{8}$	1, 1	1, 2	$\underline{9}$, 0

G^∞ :

	X	Y
A		
B		

Can't go further

IDSDS is not conclusive

Prediction: X or Y, A or B

1. Underline best-response payoffs
2. No dominant strategies. But are any *strictly dominated*?
    - Dominated $\Rightarrow$ never a best response. Only Z has no underline
3. Z is strictly dominated by _______. Delete it!
4. D's only underline was in Z. Now it is dominated by _______. Delete it!
5. W's only underline was in D. Now it is dominated by _______. Delete it!
6. C's only underline was in W. Now it is dominated by _______. Delete it!
7. Nothing left to delete: G^∞ is a 2×2 game. IDSDS is not **decisive**.

Log In Again: Pick a Number

MobLab: “Beauty contest”

Choose an integer from **1** to **100**.

Whoever is closest to $\frac{2}{3} \times$(the average of everyone’s number) wins.

We’ll play a few rounds.

Back to Your Numbers

$\frac{2}{3}$ of the average can never exceed $\frac{2}{3} \times 100 \approx 66.7$.

- So 67 strictly dominates everything above it. Delete 68–100.
- Now the target can't exceed $\frac{2}{3} \times 67 \approx 44.7$. Delete 45–67.
- Now it can't exceed $\frac{2}{3} \times 45 = 30$. Delete 31–45.
- Keep going … where does it end?

Help me: Why does IDSDS fail? Logically it makes so much sense...

IDWDS, and Why It's Trickier

What if we delete **weakly** dominated strategies too?

Player 1 \ Player 2	L	R
A	4, 0	0, 0
T	3, 2	2, 2
M	1, 1	0, 0
B	0, 0	1, 1

Two perfectly reasonable places to start:

- M is strictly dominated by T ← delete this first
- B is strictly dominated by T

IDWDS, and Why It's Trickier

What if we delete **weakly** dominated strategies too?

		Player 2: L	Player 2: R
Player 1	A	4, 0	0, 0
	T	3, 2	2, 2
	M	1, 1	0, 0
	B	0, 0	1, 1

Two perfectly reasonable places to start:

- M is strictly dominated by T
- B is strictly dominated by T $\leftarrow$ delete this first

IDWDS

Does IDWDS have the same rationality foundation?

		Steven C	Steven D
Sarah	C	50, 50	0, 100
Sarah	D	100, 0	0, 0

D is weakly dominated but is it *definitely* irrational?

What if Sarah were *positive* that Steven would play *D*?

Our Main Solution Concept:
Nash Equilibrium

Still Not Enough

Games solved outright by IDSDS or IDWDS are the exception.

We need something that applies to *every* game.

The idea: stop asking "what will a rational player do?" and start asking

"which profiles are **stable**?"

An Example

		P2	
		L	R
P1	U	1, 1	0, 0
	D	0, 0	2, 2

Profile (U, L) is **stable**:

1. Player 1 is hurt by deviating from U to D
2. Player 2 is hurt by deviating from L to R

An Example

		P2 L	P2 R
P1	U	1, 1	0, 0
P1	D	0, 0	2, 2

Profile (U, L) is **stable**:

1. Player 1 is hurt by deviating from U to D
2. Player 2 is hurt by deviating from L to R

Profile (D, R) is also **stable**:

1. Player 1 is hurt by deviating from D to U
2. Player 2 is hurt by deviating from R to L

Definition: Nash Equilibrium (Two Players)

In a 2-player game, a strategy profile $s^* = (s_1^*, s_2^*)$ is a **Nash equilibrium** if

1. for every $s_1 \in S_1$: $\pi_1(s_1^*, s_2^*) \; \geq \; \pi_1(s_1, s_2^*)$
2. for every $s_2 \in S_2$: $\pi_2(s_1^*, s_2^*) \; \geq \; \pi_2(s_1^*, s_2)$

In words: **assuming the other player sticks to the Nash equilibrium**, neither player can do better by changing their strategy.

Does not allow **joint deviations** (see previous example)

John Forbes Nash Jr. (1928–2015)

- Duffin's reference letter, 1948, in full:
  "He is a mathematical genius."
- Princeton PhD, 1950: 26 pages, two citations
- One of the two was his own
- Nobel, 1994, with Harsanyi and Selten
- Abel Prize, 2015: PDEs, not games
- Schizophrenia from 1959, three decades away
- Died May 2015, coming home from the Abel ceremony

Another Example

		Player 2 L	Player 2 C	Player 2 R
Player 1	T	3, 2	0, 0	1, 1
Player 1	M	3, 0	1, 5	4, 4
Player 1	B	1, 0	2, 3	3, 0

Verify (T, L) is a Nash equilibrium (NE):

- Player 1, holding L fixed: T gives 3; M gives 3; B gives 1 ✓
  - Nothing is **strictly** better. (Ties are okay.)
- Player 2, holding T fixed: L gives 2; C gives 0; R gives 1 ✓

So (T, L) is a Nash equilibrium. Find the other one.

Four Ways to Say It

No regret. After seeing what the other player did, nobody wishes they had chosen differently.

Self-enforcing agreement. If the players could talk beforehand and shake on s^*, nobody would want to break it, assuming the other keeps it.

Viable recommendation. If a third party publicly recommended s^*, nobody would want to deviate, assuming the others comply.

Rationality with correct beliefs. Each player best-responds to what they believe the other will do, and those beliefs are right.

Best Replies

Strategy s_i is a **best reply** to s_{-i} if

$$\pi_i(s_i, s_{-i}) \;\geq\; \pi_i(s_i', s_{-i}) \quad \text{for every } s_i' \in S_i$$

Equivalent definition of Nash equilibrium:

$$s \text{ is a Nash equilibrium} \iff \text{every } s_i \text{ is a best reply to } s_{-i}$$

This is the version worth memorizing. It also tells you how to *find* equilibria.

The Underlining Trick

1. In each **column**, underline Player 1's largest payoff (all ties)
2. In each **row**, underline Player 2's largest payoff (all ties)
3. Any cell with **both** numbers underlined is a Nash equilibrium

Player 1 \ Player 2	E	F	G	H
A	4, 0	3, 2	2, 3	4, 1
B	4, 2	2, 1	1, 2	0, 2
C	3, 6	5, 5	3, 1	5, 0
D	2, 3	3, 2	1, 2	3, 3

Your turn. Columns first, then rows.

The Underlining Trick

		Player 2			
		E	F	G	H
Player 1	A	$\underline{4}, 0$	$3, 2$	$2, \underline{3}$	$\underline{4}, 1$
	B	$\underline{4}, \underline{2}$	$2, 1$	$1, \underline{2}$	$0, \underline{2}$
	C	$3, \underline{6}$	$\underline{5}, 5$	$\underline{3}, 1$	$\underline{5}, 0$
	D	$2, \underline{3}$	$3, 2$	$1, 2$	$3, \underline{3}$

Only (B, E) has both payoffs underlined: the unique Nash equilibrium.

Watch the near-misses:

- (A, E): Player 1 is happy, Player 2 would rather move to G
- (C, F): Player 1 is happy, Player 2 would rather move to E
- (D, H): Player 2 is happy, Player 1 would rather move to C

More Than Two Players

A profile $s^* \in S$ is a Nash equilibrium if, for every player i,

$$\pi_i(s^*) \;\geq\; \pi_i(s_1^*, \ldots, s_{i-1}^*,\; s_i,\; s_{i+1}^*, \ldots, s_n^*) \quad \text{for all } s_i \in S_i$$

n inequalities, one per player. Nothing new, just more of them.

A Three-Player Example

Let's try a 3-player game ($2 \times 2 \times 2$)

Player 3: E

Player 1 \ Player 2	L	R
T	4, 1, 4	0, 0, 1
B	3, 1, 3	3, 3, 2

Player 3: F

Player 1 \ Player 2	L	R
T	0, 0, 1	2, 2, 3
B	1, 4, 2	1, 0, 3

Two equilibria in here. Find them using the underlining technique.

A Three-Player Example

Let's try a 3-player game ($2 \times 2 \times 2$)

Player 3: E

Player 1 \ Player 2	L	R
T	$\underline{4}, \underline{1}, \underline{4}$	$0, 0, 1$
B	$3, 1, \underline{3}$	$\underline{3}, \underline{3}, 2$

Player 3: F

Player 1 \ Player 2	L	R
T	$0, 0, 1$	$\underline{2}, \underline{2}, \underline{3}$
B	$\underline{1}, \underline{4}, 2$	$1, 0, \underline{3}$

Two equilibria in here. Find them using the underlining technique.

A Three-Player Example

Let's try a 3-player game ($2 \times 2 \times 2$)

Player 2

Player 1	L	R
T	$\underline{4}, \underline{1}, \underline{4}$	$0, 0, 1$
B	$3, 1, \underline{3}$	$\underline{3}, \underline{3}, 2$

Player 3: E

Player 2

Player 1	L	R
T	$0, 0, 1$	$\underline{2}, \underline{2}, \underline{3}$
B	$\underline{1}, \underline{4}, 2$	$1, 0, \underline{3}$

Player 3: F

Two equilibria in here. Find them using the underlining technique.

(T, L, E) and (T, R, F): the only cells with all three payoffs underlined.

Check (B, R, E) and (B, L, F): two underlines each. Players 1 and 2 are content; only Player 3 wants to move.

Nash Survives IDSDS

All three procedures return a **set of strategy profiles**, not one prediction:

- $NE(G)$: the Nash equilibrium profiles for game G
- $IDSDS(G)$: the profiles that survive IDSDS in game G
- $IDWDS(G)$: the profiles that survive IDWDS in game G

Each is a subset of S, and each can be empty, a single profile, or many.

Theorem. For every ordinal strategic-form game G,

$$NE(G) \subseteq IDSDS(G)$$

- A Nash equilibrium never uses a strategy that IDSDS deletes
- So IDSDS never throws away a Nash equilibrium: it's safe
- But the same is **not** true of IDWDS: it is possible that $NE(G) \cap IDWDS(G) = \emptyset$

Strict Nash Equilibrium

s^* is a **strict** Nash equilibrium if every player is *strictly* worse off deviating, rather than merely no better off.

$$\text{Nash:} \quad \pi_i(s^*) \;\geq\; \pi_i(s_i, s^*_{-i}) \quad \text{for every } s_i \in S_i$$

$$\text{Strict:} \quad \pi_i(s^*) \;>\; \pi_i(s_i, s^*_{-i}) \quad \text{for every } s_i \neq s_i^*$$

Two changes: $\geq$ becomes $>$, and s_i^* itself is excluded. It has to be, since $\pi_i(s^*) > \pi_i(s^*)$ is false.

Theorem. If $IDSDS(G) = \{s^*\}$ then s^* is a strict Nash equilibrium.

If $IDWDS(G) = \{s^*\}$ then s^* is a Nash equilibrium, but not necessarily a strict one.

A Bigger Example

Suppose OSU donor Ratmir Timashev makes the following offer to 50 students. Each secretly writes down a request: a multiple of $10, up to $100.

- If **5 or fewer** students ask for $100, everyone gets what they asked for
- Otherwise, everyone gets nothing

All students are selfish and greedy. What are the Nash equilibria?

A Bigger Example

There are two families of equilibria.

(a) Seven or more students request $100.
Everyone gets nothing. Can anyone gain by switching alone?
Is this a **strict** Nash equilibrium?

(b) Exactly five request $100, and all 45 others request $90.
Everyone is paid. Can anyone gain by switching alone?
Is this a **strict** Nash equilibrium?

Convince yourself that *exactly six* requesting $100 is not an equilibrium.

When There Is None

Matching Pennies. Both match $\Rightarrow$ Player 1 wins; otherwise Player 2 wins.

		Player 2	
		H	T
Player 1	H	$\underline{1}, 0$	$0, \underline{1}$
	T	$0, \underline{1}$	$\underline{1}, 0$

No cell has both payoffs underlined.

With only ordinal payoffs, a game need not have **any** Nash equilibrium.

Fixing this requires letting players randomize, and randomizing requires cardinal payoffs. That's Chapter 5.

Infinite Strategy Sets

No Table, Same Definition

Two players. Each writes down a real number ≥ 1, so $S_1 = S_2 = [1, \infty)$.

Player 1 writes x, Player 2 writes y:

$$\pi_1(x,y) = \begin{cases} x-1 & \text{if } x < y \\ 0 & \text{if } x \geq y \end{cases} \qquad \pi_2(x,y) = \begin{cases} y-1 & \text{if } x > y \\ 0 & \text{if } x \leq y \end{cases}$$

You want to write the larger… no, the *smaller* number. But not too small.

What are the Nash equilibria?

The Answer Is $(1, 1)$, and Only $(1, 1)$

It is an equilibrium. At $(1, 1)$ both get 0. If Player 1 switches to any $x > 1$, then $x \geq y = 1$, so they still get 0.

Nothing else is.

- If $x = y > 1$: Player 1 gets 0, but any $\hat{x}$ with $1 < \hat{x} < x$ gives $\hat{x} - 1 > 0$
- If $x < y$: Player 1 gets $x - 1$, but any $\hat{x}$ with $x < \hat{x} < y$ gives $\hat{x} - 1 > x - 1$
- If $y < x$: same argument for Player 2

And now the strange part. For Player 1, $x = 1$ is **weakly dominated** by every other strategy.

So the unique Nash equilibrium has both players using a weakly dominated strategy.

Cournot Competition

Shell and Speedway sell an identical good: a gallon of regular. Station *i* chooses quantity $q_i \geq 0$ at constant unit cost c. The price is a *function* of total output:

$$P(q_1, q_2) = a - b(q_1 + q_2), \qquad a > c,\ b > 0$$

Shell's profit:

$$\pi_1(q_1, q_2) = P(q_1, q_2)\, q_1 - cq_1 = (a - c)q_1 - bq_1^2 - bq_1q_2$$

Strategy sets are $[0, \infty)$ (infinite). But a Nash equilibrium is still just a pair $(\bar{q}_1, \bar{q}_2)$ where neither station wants to move.

Cournot wrote this down in 1838, inventing Nash equilibrium about 120 years early, for this one class of games.

Variables and Parameters

Not every letter in a model plays the same role.

Parameters: a, b, c

- fixed numbers, known to everyone, chosen by nobody
- placeholders for a number that is already settled before anyone moves
- left as letters so one solution covers every version of the game

Choice variables: q_1, q_2

- placeholders for a number that someone gets to pick
- these are the strategies: $S_i = [0, \infty)$

Derived values (functions of the variables): $P(q_1, q_2)$, $\pi_i(q_1, q_2)$

- not fixed in advance, and nobody chooses them directly
- they are *functions*: feed in the choices, get a number out

For any letter, ask: could a player have made it come out differently?

Maximizing, With Two Variables

From the math review: to maximize $f(x)$, set $f'(x) = 0$.

Shell's profit depends on *two* quantities:

$$\pi_1(q_1, q_2) = (a - c)q_1 - bq_1^2 - bq_1q_2$$

But Shell only chooses q_1. Speedway's q_2 is not Shell's to pick.

So treat q_2 as a fixed number and maximize over q_1 alone.

Partial Derivatives

A **partial derivative** is an ordinary derivative taken while every other variable is held constant, treated as though it were just a number.

$$\frac{\partial f}{\partial x} : \quad \text{differentiate in } x, \text{ treat } y \text{ as a constant}$$

The curly ∂ is the only new thing. The rules are the ones you already know.

$f(x, y) = 3x^2 + 5xy + y^3$

$$\frac{\partial f}{\partial x} = \underline{\hspace{8cm}}$$

$$\frac{\partial f}{\partial y} = \underline{\hspace{8cm}}$$

Why That Works

Same $f(x, y) = 3x^2 + 5xy + y^3$. Suppose y is stuck at 2:

$$f(x, 2) = 3x^2 + 10x + 8$$

That is a function of x alone. Differentiate it the ordinary way:

$$\frac{d}{dx} f(x, 2) = \underline{\hspace{8cm}}$$

Now compare it to $\dfrac{\partial f}{\partial x}$ with $y = 2$ substituted in.

Holding a variable constant and plugging in a constant are the same operation.

Finding the Maximizer

Two conditions, in words first:

- **First order**: at the top, the slope in the *x* direction is flat
- **Second order**: the curve bends downward there, so flat means a peak rather than a valley

$$\underbrace{\frac{\partial f}{\partial x} = 0}_{\text{slope is flat}} \qquad \text{and} \qquad \underbrace{\frac{\partial^2 f}{\partial x^2} < 0}_{\text{bending downward}}$$

Skip the second condition and a flat point could just as easily be a minimum.

Shell's Problem

$$\pi_1(q_1, q_2) = (a - c)q_1 - bq_1^2 - bq_1q_2$$

Hold q_2 fixed. First order:

$$\frac{\partial \pi_1}{\partial q_1} = \underline{\hspace{10cm}} = 0$$

Second order:

$\dfrac{\partial^2 \pi_1}{\partial q_1^2} = \underline{\hspace{5cm}}$, negative because $b > 0$, so the flat point is a maximum.

From First-Order Condition to Best Reply

Shell's first-order condition holds at Shell's optimum, *whatever* q_2 turns out to be:

$$a - c - 2bq_1 - bq_2 = 0$$

So solve it for q_1:

$$BR_1(q_2) = \frac{a - c - bq_2}{2b}$$

The second-order condition was $-2b < 0$, so this really is Shell's best, not its worst.

A Best Reply Is a Function

$$BR_1(q_2) = \frac{a - c - bq_2}{2b}$$

Feed it a quantity for Speedway, and it hands back Shell's best answer to that quantity. One input, one output.

- If Speedway floods the market, BR_1 tells Shell to pull back
- If Speedway sits out, BR_1 tells Shell to act like a monopolist

Speedway's problem is the same with the labels swapped:

$$BR_2(q_1) = \frac{a - c - bq_1}{2b}$$

$BR_i(q_{-i})$ is truncated at 0 if needed.

Nash Equilibrium Is a Fixed Point

Recall the best-reply version of the definition: (q_1^*, q_2^*) is a Nash equilibrium exactly when each is a best reply to the other.

$$q_1^* = BR_1(q_2^*) \qquad \text{and} \qquad q_2^* = BR_2(q_1^*)$$

Substitute the second into the first:

$$q_1^* = BR_1\big(BR_2(q_1^*)\big)$$

Two equations in two unknowns just became **one** equation in one unknown.

Start at q_1^*, ask what Speedway would do, then ask what Shell would do back. A Nash equilibrium is a quantity that returns to *itself* after that round trip.

Solving the Fixed Point

$$q_1 = \frac{a - c - b\,BR_2(q_1)}{2b}, \qquad BR_2(q_1) = \frac{a - c - bq_1}{2b}$$

Substitute, then solve for q_1:

And Then the Other One

Feed q_1^* back through Speedway's best reply:

$$q_2^* = BR_2(q_1^*) = \frac{a - c - b\left(\frac{a-c}{3b}\right)}{2b}$$

$q_2^* =$ ____________________________

No surprise that it matches q_1^*: the two stations face identical problems, so the equilibrium had to be symmetric.

Worth checking directly that $BR_1(q_2^*) = q_1^*$. It does.

Cournot: The Numbers

At $\bar{q}_1 = \bar{q}_2 = \dfrac{a-c}{3b}$:

- Price: $$P(\bar{q}_1, \bar{q}_2) = \frac{a+2c}{3}$$
- Each profit: $$\frac{(a-c)^2}{9b}$$

Try it: $a = 25,\ b = 2,\ c = 1.$

$\bar{q}_1 = \bar{q}_2 =$ __________ $P =$ __________ profit $=$ __________

Let’s Work One Out

Three Bars on High Street

Ann, Ben, and Cal each run a bar. Before a home game each independently commits to a **Big** promotion or a **Small** one.

Cal: Big

Ann \ Ben	Big	Small
Big	2, 2, 2	0, 3, 1
Small	3, 0, 1	1, 1, 0

Cal: Small

Ann \ Ben	Big	Small
Big	1, 1, 3	0, 0, 2
Small	0, 2, 0	2, 2, 3

Ann picks the row, Ben the column, Cal the table. Payoffs in order: Ann, Ben, Cal.

Three Bars on High Street

1. Does anyone have a dominant strategy?
2. Does IDSDS delete anything at all?
3. Find every Nash equilibrium. Underline Ann's best payoff in each column, Ben's in each row, and Cal's whenever it beats her payoff in the same cell of the other table.
4. Is either equilibrium Pareto superior to the other?

What Have We Learned, Charlie Brown?

- **Second-price auction**: truthful bidding is weakly dominant (Vickrey)
- **Pivotal mechanism**: truthful reporting is weakly dominant (Clarke)
- **IDSDS**: delete strictly dominated strategies, repeat (order-free, justified by common knowledge of rationality)
- **IDWDS**: order matters, so delete simultaneously, and the justification is murkier
- **Nash equilibrium**: nobody gains by deviating alone
- $NE \subseteq IDSDS$, but a game may have no Nash equilibrium at all

Next: what happens when players can randomize.

Reading & Practice

- Bonanno, *Game Theory*, §2.3–2.7
- Exercises: §2.9.3 (auctions), §2.9.4 (pivotal mechanism), §2.9.5 (iterated deletion), §2.9.6 (Nash equilibrium), §2.9.7 (infinite games)
- Proofs of both theorems are in §2.8
- Solutions in §2.10. Try them first